MacHuna v1.6.12 — User Manual

Broadcast Media Format Converter

DNS Vision Limited — For Vision Mixers, TDs, and Support Engineers

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1. Overview

MacHuna is a macOS application for translating broadcast media assets between formats. It converts video clips, TGA sequences, and still images to Grass Valley Kahuna `.SWS` format, reads and writes Grass Valley Kayenne `.EIF` native clips, and extracts `.SWS` files back to standard formats for use on Kayenne and Sony MVS desks.

It is a Mac-native alternative to the Windows-only K-Watch application included with Grass Valley K-Manager Pro.

1.1 What MacHuna Does

Direction	From	To
Kahuna SWS	MOV, MP4, MXF, MKV, AVI, TGA sequences, PNG, BMP, JPG	`.SWS` (Grass Valley Kahuna)
Kayenne EIF	MOV, MP4, MXF, TGA sequences, `.SWS`	`.EIF` (Grass Valley Kayenne) *(UNCONFIRMED on hardware)*

Direction	From	To
Kayenne EIF → SWS	`.EIF`	`.SWS` (Grass Valley Kahuna) *(UNCONFIRMED on hardware)*
Kayenne EIF → TGA	`.EIF`	32-bit RGBA TGA sequence *(UNCONFIRMED on hardware)*
Kayenne TGA	`.SWS` , `.EIF`	32-bit RGBA TGA sequence *(UNCONFIRMED on hardware)*
Sony TGA	`.SWS` , `.EIF` , TGA sequences	32-bit RGBA TGA sequence (Sony MVS naming)
TGA Sequence	TGA sequences, video clips	32-bit RGBA TGA sequence (i⍰p standards conversion)

1.2 Supported Standards

All nine standards below have been confirmed against K-Watch reference files and verified on a live Grass Valley Kahuna mainframe.

Standard	fps	Region
1080i/50	25	UK / Europe
1080i/59.94	29.97	USA / Japan
1080i/60	30	—
1080p/25	25	UK / Europe
1080p/50	50	UK / Europe
1080p/59.94	59.94	USA / Japan
1080p/60	60	—

1.3 Key Differences from K-Watch

- Runs natively on macOS — no Windows, no Parallels
- Accepts a wider range of input formats (K-Watch supports MOV and AVI only)
- Converts between Kahuna SWS, Kayenne EIF, and Sony MVS formats in one app
- Reads and writes Kayenne `.EIF` native clips — format reverse-engineered from real Kayenne hardware
- Built-in Video Player for checking `.SWS` and `.EIF` files without a Kahuna or Kayenne desk

> **NOTE** MacHuna replicates K-Watch's conversion functionality. It does not replicate K-Manager Pro's network upload to mainframe or project synchronisation features.

2. Installation

MacHuna is distributed as a self-contained `.app` bundle. ffmpeg is bundled inside — no separate installation required.

2.1 First Launch

1. Copy `MacHuna.app` to your Applications folder, or run it from any location
2. On first launch, right-click the app and choose **Open** to bypass macOS Gatekeeper
3. Subsequent launches work by double-clicking normally

> **NOTE** Gatekeeper may warn that the app is from an unidentified developer. This is expected — MacHuna is not signed with an Apple Developer certificate. Right-click → Open bypasses this check.

2.2 Settings

MacHuna saves all settings automatically on quit. Settings are stored at:

```
~/.kwatch_settings.json
```

To reset to defaults, quit MacHuna and delete this file.

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3. Main Window

The MacHuna window has three rows:

Row	Purpose
Destination Folder	Where converted files are written
Convert	Open files, choose output format, convert
Log	Conversion progress and errors

The Convert row adapts automatically based on what files you open. MacHuna detects the input type and shows only the controls that are relevant.

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4. Converting Files

The workflow is the same regardless of what you are converting or what you are converting it to:

1. Set your **Destination Folder**
2. Click **Open Files…** and select a folder

3. MacHuna scans the folder, detects the input type, and populates the file list
4. Choose your **Output** format from the dropdown
5. Set any options that appear (standard, clip name, field order, etc.)
6. Click **Convert**

A conversion log is written to the Destination Folder when the batch completes.

4.1 Input Type Detection

MacHuna reads the contents of the folder you open and determines the input type automatically:

Files found in folder	Detected as	Available outputs
`.SWS` files only	SWS source	Kahuna SWS, Kayenne TGA, Kayenne EIF, Sony TGA
`.EIF` files only	EIF source	Kahuna SWS, Kayenne TGA, Sony TGA
Mix of `.EIF` and `.SWS` files	EIF + SWS source	Kahuna SWS, Kayenne TGA, Sony TGA
Video files only (MOV, MP4, MXF…)	Video source	Kahuna SWS, Kayenne TGA, Kayenne EIF, Sony TGA, TGA Sequence
TGA sequences and/or stills	TGA / stills source	Kahuna SWS, Kayenne EIF, Sony TGA, TGA Sequence — see note below
Mix of SWS and other files	Error — shown in summary	—

> **NOTE** If you see a "mixed input" error, the folder contains both `.SWS` files and other file types. Move them into separate folders and convert each folder independently.

> **NOTE — stills go to Kahuna SWS only.** A still image is a single frame, not a clip, so MacHuna does not convert stills to Kayenne EIF, Sony TGA or TGA Sequence output. Those outputs need a clip or a TGA sequence. If a still is selected with one of them, MacHuna names the file and stops rather than converting it. Deselect the still, or choose Kahuna SWS. A folder holding both stills and TGA sequences is fine: select the sequence and leave the stills unticked.

4.2 The File List

The file list shows one entry per item to be converted:

- **TGA sequences** are collapsed to a single line: `TNTS201 (30 frames)` — you do not need to select individual frames
- **Video files** appear by filename
- **Stills** appear by filename
- **EIF files** appear by filename

Click entries to select or deselect them. By default, all files are selected.

4.3 Numbering

For Kahuna SWS output, the **Start Number** field sets the number assigned to the first output file. Subsequent files are numbered sequentially.

Use source file number — tick this when re-converting K-Watch named files (e.g. `TNTS201_30_0001.tga`). MacHuna reads the slot number from the filename rather than the Start Number. Useful when a conversion needs to replace a specific existing slot on the Kahuna.

For Kayenne EIF output, a **Start slot** spinner sets the first slot number (e.g. `0001`). Output files are named `0001.eif`, `0002.eif`, and so on.

4.4 Cancel

Click **Cancel** during a conversion to stop after the current file. The conversion log is not written if cancelled mid-batch.

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5. Kahuna SWS Output

When outputting to Kahuna SWS, the following options are available:

Option	Description
Standard	Target video standard. 1080p/50 is most common for UK/European broadcast.
Split >4GB	On by default. Files over 4GB are split into 2GB FAT32-safe chunks. See Section 11.
Ignore alpha	No key plane is written. Output is fill-only, matching K-Watch no-alpha behaviour. Use for fill-only content or when the Kahuna output does not use a downstream keyer.
Auto play	Sets the Auto Play flag in the SWS header. The Kahuna begins playback when the clip is loaded.
Loop play	Sets the Loop Play flag in the SWS header. The Kahuna loops the clip continuously.
TGA source interlaced	See Section 8.2.
Include audio	Embeds audio from the source into the .SWS file. Shown only when audio is detected in the source. See Section 9.

> **SWS → SWS standards conversion:** when the source is itself an SWS file, the output header's clip name follows the source SWS's name, and the output key state follows the source (a keyless source produces keyless output). **Audio is not carried through an SWS → SWS conversion** — if the source SWS contains embedded audio, it is dropped and a warning is written to the log.

5.1 Progressive to Interlaced (P→I)

When converting a progressive source to an interlaced standard, MacHuna field-weaves pairs of progressive frames into genuine interlaced frames. The frame count halves — a 50fps progressive source becomes 25fps interlaced output. This is the correct behaviour for the Kahuna.

Example: 100 frames of 1080p/50 source → 50 frames of 1080i/50 output.

The source must run at the interlaced field rate (double the frame rate). Weaving pairs of frames only keeps the right duration when the source is a genuine field-rate stream:

Progressive source	Interlaced target	Result
50p	1080i/50	✅ correct — weaves to 25 interlaced frames
59.94p	1080i/59.94	✅ correct
60p	1080i/60	✅ correct
25p	1080i/50	⛔ blocked — would play at 2× speed
29.97p	1080i/59.94	⛔ blocked
30p	1080i/60	⛔ blocked

If you give MacHuna a **same-rate** progressive source (e.g. a 25p file destined for 1080i/50) or any other incompatible rate, it will **stop with a clear error and write no file**, rather than silently produce a clip that plays at the wrong speed. Convert the source to the correct field rate first (e.g. 25p → 50p), or choose a progressive output standard. *(This rate check applies to video-clip, SWS→SWS and clip→TGA conversions, which carry a known frame rate. A loose TGA image sequence has no frame rate to check and is still assumed to be a double-rate field stream.)*

5.2 Interlaced to Progressive (I→P)

When converting an interlaced source to a progressive standard, MacHuna bob-deinterlaces to produce the correct number of progressive frames. Playback speed on the Kahuna is preserved.

Example: 50 frames of 1080i/50 source → 100 frames of 1080p/50 output.

Cross-rate conversions (corrected in v1.6.12). Deinterlacing doubles the frame count, which lands on the target rate exactly only when the progressive standard is double the interlaced source's frame rate — 1080i/50 → 1080p/50, or 1080i/59.94 → 1080p/59.94. Any other pairing needs a rate change on top, and before v1.6.12 MacHuna did not apply one, so the clip played fast or slow:

Source	Target	Before v1.6.12	Now
1080i/50	1080p/50	correct	correct
1080i/50	1080p/60	ran fast	correct
1080i/59.94	1080p/50	ran fast	correct
1080i/59.94	1080p/25	ran 20% slow	correct

This applies to video clips, SWS→SWS, and TGA outputs. **A loose TGA image sequence is the exception:** it carries no frame rate anywhere, so MacHuna assumes the source standard matches the output family you pick (an interlaced pile aimed at 1080p/50 is treated as 1080i/50 material). That assumption is right for normal use. If you are converting a TGA sequence across families — interlaced 50Hz frames aimed at a 60Hz progressive standard — convert via a video clip or SWS instead, so MacHuna has a real frame rate to work from.

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6. Kayenne EIF — Native Kayenne Format

> **IMPORTANT — Hardware Status** EIF write and conversion paths have been verified correct by analysis against real Kayenne-produced reference files, but **none have been tested on a live Kayenne desk**. MacHuna will warn you before converting to or from EIF. Verify the first import carefully.

MacHuna can read and write Grass Valley Kayenne `.EIF` files — the native clip format used by Kayenne ClipStore and Image Store. The format was fully reverse-engineered from real Kayenne-produced files.

6.1 What is EIF?

`.EIF` is the Kayenne's native clip container. Each file holds a complete clip — fill and key combined in a single proprietary pixel format. The format stores 1920×1080 progressive video only; frame rates are either 25fps or 50fps. Files are named by slot number: `0001.eif`, `0002.eif`, and so on.

6.2 Converting TO EIF

Any of the following inputs can be converted to Kayenne EIF:

Input	Notes
MOV, MP4, MXF, MKV, AVI	Alpha channel (fill + key) preserved if present
TGA sequence	Progressive or interlaced source — see TGA source interlaced option
Kahuna SWS	Lossless direct YCbCr repack — no RGB conversion

Output options:

Option	Description
Start slot	First slot number for output files. Files are named `0001.eif`, `0002.eif` etc. and increment per item in the batch.
TGA source interlaced	Shown when source is a TGA sequence. Tick when TGA frames are from an interlaced source. MacHuna deinterlaces each frame using yadif (field separation, TFF) to produce 50fps progressive EIF output.

EIF output is always 1920×1080. Sources of other sizes are scaled. Frame rate is rounded to the nearest EIF-supported rate (25fps or 50fps), and the video is resampled to that rate so the clip keeps its original duration and plays at the correct speed — for example a 60fps source becomes a 50fps EIF clip of the same length.

6.3 Converting FROM EIF

EIF files can be converted to the following outputs:

Output	Notes
Kahuna SWS	Lossless direct YCbCr repack. Output standard auto-derived from EIF frame rate (25fps → 1080p/25, 50fps → 1080p/50).
Kayenne TGA	Full-resolution 1920×1080 32-bit RGBA TGA sequence. Frames numbered `0001.tga` onwards.
Sony TGA	32-bit RGBA TGA sequence with 4-character clip name prefix.

For TGA outputs, select the **Standard** to control whether MacHuna field-weaves frames (interlaced standard) or extracts them as-is (progressive standard). The same field order and clip name options as regular extraction apply — see Section 7.

6.4 EIF in the Video Player

The built-in Video Player opens `.EIF` files directly. Frame rate is detected automatically from the file header — no prompt required. Fill, key, and composite panels are all populated. See Section 10.

7. Extraction Outputs — Kayenne and Sony MVS

> **IMPORTANT — Hardware Status** The extraction output paths have been confirmed correct by code analysis. However, Kayenne TGA output has **never been loaded on a live Kayenne desk**, and Sony TGA clip naming has **never been verified on a live Sony MVS**. MacHuna will warn you before converting to these targets. Use with that caveat in mind and verify the first import on your desk carefully.
>
> *(The former Kayenne MOV output was withdrawn in v1.6.5 — it was never confirmed on hardware and could not be offered consistently across input types.)*

7.1 Output Targets

Output	Format	Destination desk
Kayenne TGA	32-bit RGBA TGA sequence. Frames numbered `0001.tga` onwards. One subfolder per SWS.	Grass Valley Kayenne Image Store

Output	Format	Destination desk
Sony TGA	32-bit RGBA TGA sequence. Frames numbered `XXXX0000.tga` (4-character clip name + frame number). One subfolder per SWS.	Sony MVS Image Store

7.2 Workflow

1. Open a folder of `.SWS` files
2. Select the output target from the **Output** dropdown
3. Set any options shown (Standard, Clip name, Field order, Include audio)
4. Click **Convert**

7.3 Standard (TGA outputs)

For Kayenne TGA and Sony TGA, select the **Standard** matching your target desk's video standard. This determines whether MacHuna applies field-weaving (for interlaced output standards) or extracts frames as-is (for progressive standards).

- **Progressive standard selected:** frames extracted directly from the SWS
- **Interlaced standard selected:** if the source SWS is already interlaced, frames are passed through. If the source is progressive, pairs of frames are field-woven into interlaced output.

MacHuna logs a note describing what it did for each file.

7.4 Field Order (TGA outputs)

A **TFF / BFF** toggle appears for interlaced standards, and always for Sony TGA. **TFF (Top Field First) is the default** — it is the SMPTE standard for HD (including 1080i) and is correct for all known 1080i HD workflows (default since v1.6.3). If you see motion artefacts or comb effects on the desk after import, switch to BFF and reconvert.

> **Fixed in v1.6.12.** For Sony TGA output from a TGA sequence or a video clip, this toggle was displayed but had no effect — the conversion was hardcoded to TFF, so switching to BFF changed nothing in the output. It now works in both conversion directions, and the conversion log states which field order was applied, so you can confirm what was used. If you tested Sony TGA output before v1.6.12 and concluded BFF did not help, that test was not valid and is worth repeating.

7.5 Sony TGA — Clip Name

Enter a **4-character alphanumeric clip name** (e.g. `WIPE`). All TGA frames in the batch share this name — on the Sony MVS, files with the same 4-character prefix are grouped into a single clip on import.

If you are converting multiple distinct clips in one batch, be aware they will all share the same clip name. Convert each clip separately if they need to appear as separate clips on the desk.

7.6 Output Structure

Output	File naming
Kayenne TGA	Subfolder per SWS, named after the SWS stem. Frames `0001.tga` … inside.
Sony TGA	Subfolder per SWS, named after the 4-character clip name. Frames `XXXX0000.tga` … inside.

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8. TGA Sequences

TGA sequences are multi-frame clips stored as individually numbered still images. MacHuna handles them through the folder browser — you do not need to select individual frames.

MacHuna accepts any numbered TGA naming convention — K-Watch names, After Effects exports, third-party renders, or anything else:

```
TNTS201_30_0001.tga    (K-Watch naming)
FEDX0000.tga           (letters + digits, no separator)
shot_0001.tga          (underscore separator)
render.0001.tga        (dot separator)
```

The only requirement is that frames are numbered sequentially.

8.1 In the Folder Browser

When you open a folder containing a TGA sequence, MacHuna collapses the entire sequence to a single entry:

```
TNTS201  (30 frames)
```

Select this entry and convert as normal. MacHuna handles the frame ordering automatically.

8.2 TGA Source Already Interlaced

If you are re-wrapping TGA frames that were previously extracted from an interlaced SWS (e.g. via MacHuna's extraction outputs), tick **TGA source interlaced.** This tells MacHuna to pass frames through directly without applying field-weaving. Without this, MacHuna would incorrectly treat the already-interlaced frames as progressive and weave them again.

When **TGA source interlaced** is ticked and you select a **progressive** target standard (e.g. 1080p/50), MacHuna deinterlaces the frames using yadif (`send_field` , TFF per SMPTE 274M) rather than duplicating them — each interlaced frame is separated into two progressive fields, doubling the frame count with correct, smooth motion. Any alpha/key channel is deinterlaced with the identical filter so fill and key stay aligned.

8.3 TGA Sequence Output

When the input is a TGA sequence or a video clip, **TGA Sequence** appears as an output option. This re-writes the frames as a new numbered TGA sequence in a subfolder, applying an interlaced?progressive conversion according to the selected **Standard** — useful for converting a sequence between field standards without going through SWS. Progressive→interlaced uses field-weaving (`tinterlace`); interlaced→progressive uses yadif deinterlacing. Both follow the **Field order** toggle where it is shown, and default to TFF otherwise. The alpha/key channel is preserved throughout.

8.3 Alpha Channel

- If TGA files have an alpha channel, the key plane is extracted automatically
- If no alpha is present and Ignore alpha is off, a solid white key plane is generated
- If Ignore alpha is ticked, no key plane is written regardless

8.4 Audio

Audio is not supported in TGA sequence conversions.

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9. Audio

MacHuna extracts audio from source files and embeds it in the `.SWS` file in K-Watch native format.

9.1 Audio Format

Encoding	16-bit signed little-endian PCM
Sample rate	48,000 Hz
Channels	16 channels interleaved
Channel mapping	Ch1 = Left, Ch3 = Right, all others silent
Samples per frame	48000 ÷ fps (e.g. 960 samples at 50fps, 1920 at 25fps)

The channel mapping (L=Ch1, R=Ch3) matches the K-Watch convention, confirmed by hex analysis of K-Watch reference files. MacHuna uses an explicit ffmpeg pan filter — a straight `-ac 16` upmix does not produce the correct layout.

9.2 Notes

- Audio is appended after the key plane in the `.SWS` file
- If the source has no audio and Include audio is ticked, the option is simply ignored — no error
- Audio detection in third-party `.SWS` files uses the audio offset and format flag fields (0x1E8 and 0x1EC). The audio frame size field at 0x1C2 is unreliable across workflows and is not used for detection

10. Video Player

The built-in Video Player lets you check a file without needing a Kahuna or Kayenne desk. Click the **Video Player** button to open it.

10.1 Accepted Inputs

Input	Fill / Key / Composite	Audio
`.SWS`	Yes	Yes (if present in file)
`.EIF` (Kayenne native)	Yes — fill, key, and composite all populated	No
`.TGA` sequence (pick any frame)	Yes (if TGAs have alpha)	No
`.MOV` , `.MP4` , `.MXF` , `.MKV` , `.AVI`	Yes (alpha preserved for ProRes 4444 etc.)	Yes

Click **Open…** and select a file. For TGA sequences, pick any frame from the sequence — MacHuna loads the whole sequence. When opening a TGA sequence, you will be prompted for the frame rate (25fps default). For EIF files, the frame rate is detected automatically from the header.

10.2 Display Layout

Panel	Content
Fill	The video content plane
Key	The alpha/key plane (greyscale)
Composite	Fill composited over a chequerboard using the key as alpha — shows transparency
Audio	VU meters for left and right channels

10.3 Transport Controls

Control	Action
Cue	Jump to the first frame
Play	Begin playback at the clip's native frame rate
Pause	Pause at the current frame
Stop	Stop and return to the first frame

10.4 Notes

- Multiple player windows can be open simultaneously
- Split `.SWS` files (>4GB, multi-chunk) cannot currently be previewed

- Audio playback uses the sounddevice library. If not installed, the player opens without audio but VU meters are still drawn

11. Large File Support (>4GB)

Files larger than 4GB are automatically split into 2GB chunks when **Split >4GB** is enabled (on by default). The split format exactly matches K-Watch output and has been confirmed working on a live Kahuna mainframe.

11.1 Split File Structure

```
201.SWS/
  01_OF_03._XX   (512-byte header + video data, exactly 2GB)
  02_OF_03._XX   (video data, exactly 2GB)
  03_OF_03._XX   (video data, remainder)
```

- Chunk 1 contains the SWS header followed by video data
- Subsequent chunks contain raw video data only
- The header in chunk 1 carries the total frame count across all chunks
- Audio is not supported in split files

> **IMPORTANT** When transferring to a Kahuna via USB, copy the entire `.SWS` folder (e.g. `201.SWS/`), not the individual chunk files inside it.

12. SWS Format Reference

This section is for support engineers and developers. It documents the SWS binary format as reverse-engineered from K-Watch reference files and verified against a live Grass Valley Kahuna mainframe.

12.1 File Layout

Range	Content
`0x000 – 0x1FF`	512-byte header (big-endian)
`0x200 – N`	Fill plane: v210 big-endian, `plane_size × frame_count` bytes
`N – M`	Key plane: v210 big-endian, same size as fill plane (absent if `play_count == 0`)
`M – EOF`	Audio data: 16-bit LE PCM, 16ch, 48kHz (absent if audio offset == 0)

12.2 Key Header Fields

Offset	Type	Description
`0x188`	uint32 BE	Video standard code (OR'd with playback flags — see 12.4)
`0x18C`	uint32 BE	Format variant code (unambiguous fps lookup — all nine values are unique)
`0x190`	uint32 BE	Width in pixels
`0x194`	uint32 BE	Height in pixels (fill plane)
`0x1A0`	uint32 BE	Plane size (bytes per frame, fill or key)
`0x1A4`	uint32 BE	Frame count
`0x1A8`	uint32 BE	Play count (= frame count; 0 if no key plane)
`0x1B4`	uint32 BE	`(plane_size × frame_count + 512) / 32` ; 0 if no key
`0x1C2`	uint16 BE	Audio frame size — unreliable, do not use for detection
`0x1CC`	uint32 BE	Total file size in bytes
`0x1E8`	uint32 BE	Audio data offset ÷ 32 (0 if no audio)
`0x1EC`	uint32 BE	Audio format flag: `0x03000000` if audio present

12.3 Audio Detection

Reliable method: `aud_offset (0x1E8) > 0` AND `aud_fmt (0x1EC) == 0x03000000` .

Do not rely on the audio frame size field at `0x1C2` . K-Watch writes `0x1680` , but third-party tools may write different values. MacHuna uses the offset and format flag fields for all audio detection.

12.4 Playback Flags

Bits 2 (`0x04`) and 3 (`0x08`) of the low byte at `0x188` are OR'd into the video standard code:

Flag	Bit	Value
Auto Play	2	`0x04`
Loop Play	3	`0x08`

12.5 Interlaced Flag

Bit 15 (`0x8000`) of `0x188` is set for all interlaced standards. The standard code for all interlaced formats is `0xC923` .

13. Troubleshooting

Kahuna showing black key / no key

- If **Ignore alpha** is ticked, no key plane is written — correct behaviour
- If the source has no alpha channel and Ignore alpha is off, a solid white key is generated automatically
- If you are expecting a real key but getting white, check your source file has a valid alpha channel

Audio not playing on Kahuna

- Confirm **Include audio** is ticked
- Confirm the source file has an audio track (MacHuna hides Include audio if no audio is detected)
- Audio is resampled to 48kHz by ffmpeg if the source is at a different rate — this is automatic

File >4GB not loading on Kahuna

- Confirm **Split >4GB** is enabled
- Copy the entire `.SWS` folder to the USB drive, not the individual chunk files inside it
- The USB drive must be FAT32-formatted

Conversion fails with ffmpeg error

- Check the Log area for the ffmpeg error message
- Confirm the source file is not corrupted — try opening it in another application
- Confirm the Destination Folder path exists and is writable

Extraction output not loading on Kayenne or Sony MVS

- Note that Kayenne TGA output has not been confirmed on a live Kayenne desk
- Sony TGA clip naming has not been confirmed on a live Sony MVS
- Check that the correct Standard is selected for the target desk's video format
- For Sony TGA: confirm the 4-character clip name matches your expected import workflow
- For interlaced output: if motion artefacts appear, try switching the Field Order toggle (TFF ⍰ BFF; TFF is the default)

EIF file not loading on Kayenne desk

- EIF write output has not yet been tested on a live Kayenne desk — proceed with caution
- Confirm the file is named with the correct 4-digit slot number (e.g. `0001.eif`)
- Confirm the destination folder or USB drive is formatted correctly for Kayenne
- Check the Log area for any conversion warnings

TGA sequence not appearing as a single entry in the file list

- Confirm the TGA files are in the same folder with no other file types present

- MacHuna detects TGA sequences by parsing numbered filenames — files must be numbered sequentially. Any naming convention works (K-Watch, After Effects, custom renders, etc.) as long as the base name is consistent and frames are numbered.

File plays at double speed on Kahuna

This can happen if:

- A progressive source was converted to a Kahuna SWS without using the progressive standard, and there is a mismatch between the frame count and the interlaced header
- A TGA sequence from an interlaced source was converted without ticking **TGA source interlaced** — MacHuna field-weaved the already-interlaced frames, halving the frame count again

Check the conversion log for any warnings about P→I or interlaced detection.

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14. Known Limitations

- **Apple Silicon only** — Intel Mac builds are not supported
- **Split .SWS files** cannot be previewed in the Video Player
- **TGA sequence audio** is not supported
- **HLG Rec.2020** colour space is not implemented (requires a reference HLG .SWS file to reverse-engineer the header values)
- **EIF write and conversion** — coded and working by analysis, but not yet confirmed on a live Kayenne desk. MacHuna warns before converting to or from EIF. Verify your first import carefully.
- **EIF audio** — companion `.eaf` audio files used by some Kayenne clips are not currently supported. EIF files are always loaded without audio in the Video Player.
- **Extraction outputs unconfirmed on hardware** — Kayenne TGA and Sony MVS TGA naming have not been verified on live desks. MacHuna will warn you before converting to these targets.
- **Kayenne MOV output withdrawn** (v1.6.5) — it was never confirmed on hardware; use Kayenne TGA or Kayenne EIF instead.

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15. Authors

David Steer / DNS Vision Limited & Claude (Anthropic)

MacHuna was built collaboratively using AI-assisted development. The SWS format was reverse-engineered from K-Watch reference files and verified against a live Grass Valley Kahuna mainframe. The Kayenne EIF format was reverse-engineered from real Kayenne-produced clips.